

The sharp uniform maximum estimate in dimension three

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Abstract

For measurable symmetric coefficient fields of trace one and ellipticity lower bound δ , the uniform L^p -to-maximum estimate on the unit ball in $\mathbb{R}^3$ holds exactly for $p > \max\{3/2, 3(1 - 2\delta)\}$. The proof combines a directional Hessian integral inequality with an obstacle envelope when $\delta > 1/6$, and a shifted determinant inequality with a positive adjoint correction when $\delta \leq 1/6$. Smooth radial examples exclude both critical endpoints.

1 Statement and notation

Theorem 1.1. *Fix $0 < \delta < 1/3$ and $1 \leq p < \infty$. There is a finite constant $C = C(\delta, p) > 0$ such that*

$$\max_{x \in B_1} |u(x)| \leq C \left(\int_{B_1} |L_A u(x)|^p dx \right)^{1/p} \quad (1)$$

for every Lebesgue-measurable symmetric matrix field $A = (a_{ij})$ with

$$\operatorname{tr} A(x) = 1, \quad \xi^T A(x) \xi \geq \delta |\xi|^2 \quad \text{for every } \xi \in \mathbb{R}^3, \text{ for almost every } x \in B_1,$$

and every real-valued $u \in C^2(B_1) \cap C(\overline{B_1})$ with $u|_{\partial B_1} = 0$ and $L_A u \in L^p(B_1)$, if and only if

$$p > p_c(\delta) := \max\{3/2, 3(1 - 2\delta)\}. \quad (2)$$

Here $B_1 = \{x \in \mathbb{R}^3 : |x| < 1\}$ and $L_A u = A : D^2 u = \sum_{i,j=1}^3 a_{ij} \partial_{ij} u$.

All matrix norms without a subscript are Frobenius norms. Spherical averages are normalized to have total mass one. For a symmetric matrix H put

$$P_\delta(H) = \delta \operatorname{tr} H + (1 - 3\delta) \lambda_{\min}(H) = \min_{\substack{A \geq \delta I \\ \operatorname{tr} A = 1}} A : H. \quad (3)$$

The equality follows by writing $A = \delta I + (1 - 3\delta)B$ with $B \geq 0$ and $\operatorname{tr} B = 1$. In particular $P_\delta(H) \leq A : H$ for every admissible A . Constants below need not be optimal, but are independent of the coefficient field and of the function in (1).

2 Necessity, including the endpoints

Set $a = 1 - 2\delta$ and $b = 2\delta/a$. For $x \neq 0$ let

$$A_\delta(x) = \delta I + (1 - 3\delta) \frac{x \otimes x}{|x|^2}, \quad A_\delta(0) = I/3.$$

Its eigenvalues are (a, δ, δ) , so it is admissible. If $u(x) = v(|x|)$, then

$$L_{A_\delta} u = a \left(v'' + \frac{b}{r} v' \right).$$

For any nonnegative $f \in C_c^\infty((0, 1))$ define

$$v(r) = \frac{1}{a} \int_r^1 s^{-b} \left(\int_0^s t^b f(t) dt \right) ds. \quad (4)$$

Then $v \geq 0$, $v' \leq 0$, $v(1) = 0$ and $L_{A_\delta} u = -f(|x|)$. The function v is constant near zero; hence this u is smooth in B_1 and continuous on its closure. Every forcing norm is finite. Fubini gives

$$v(0) = \frac{1}{a} \int_0^1 t^b f(t) J_b(t) dt, \quad J_b(t) = \int_t^1 s^{-b} ds, \quad \|L_{A_\delta} u\|_p^p = 4\pi \int_0^1 f(t)^p t^2 dt. \quad (5)$$

If $\delta < 1/4$, then $b < 1$ and $J_b(t)$ is bounded above and below by positive constants on $(0, 1/2]$. Fix a nonzero nonnegative $\varphi \in C_c^\infty((1, 2))$. For $p < 3a = 3/(b+1)$, take

$$f_\varepsilon(t) = \varepsilon^{-(b+1)} \varphi(t/\varepsilon), \quad 0 < \varepsilon < 1/4.$$

Equations (4)–(5) give $v_\varepsilon(0) \geq c > 0$, while

$$\|L_{A_\delta} u_\varepsilon\|_p = C_{\varphi, p} \varepsilon^{3/p-(b+1)} \rightarrow 0.$$

At $p = 3a$, choose smooth cutoffs $0 \leq \chi, \eta \leq 1$ with $\chi(s) = 0$ for $s \leq 1$, $\chi(s) = 1$ for $s \geq 2$, $\eta(t) = 1$ for $t \leq 1/4$, and $\eta(t) = 0$ for $t \geq 1/2$. Use

$$f_\varepsilon(t) = t^{-(b+1)} \chi(t/\varepsilon) \eta(t).$$

For each ε this is an admissible smooth compactly supported source, and

$$v_\varepsilon(0) \geq c \log(1/\varepsilon) - C, \quad \|L_{A_\delta} u_\varepsilon\|_p \leq C(\log(1/\varepsilon))^{1/p}.$$

Since $p = 3a > 3/2 > 1$, the ratio tends to infinity.

For every $0 < \delta < 1/3$, the isotropic field $A = I/3$ is also admissible. In (4) take $a = 1/3$, $b = 2$; then

$$v(0) = 3 \int_0^1 (t - t^2) f(t) dt.$$

For $p < 3/2$ the choice $f_\varepsilon(t) = \varepsilon^{-2} \varphi(t/\varepsilon)$ gives a positive lower bound for $v_\varepsilon(0)$ and a forcing norm tending to zero. At $p = 3/2$, the choice $f_\varepsilon(t) = t^{-2} \chi(t/\varepsilon) \eta(t)$ gives

$$v_\varepsilon(0) \geq c \log(1/\varepsilon) - C, \quad \|L_A u_\varepsilon\|_{3/2} \leq C(\log(1/\varepsilon))^{2/3}.$$

These constructions exclude every $p \leq p_c(\delta)$. They use individual smooth functions, not singular limiting functions.

3 The estimate for $\delta > 1/6$

We prove the estimate in this range by a homogeneous Hessian calibration and a Laplacian obstacle construction. The calibration exponent will be denoted by s ; it need not equal the exponent in the asserted estimate.

3.1 A scalar projection inequality

For $h, v \in \mathbb{R}^3$, set

$$a = \frac{|h+v|}{2}, \quad b = \frac{|h-v|}{2}, \quad L(h, v) = \begin{cases} h \cdot v, & a+b \leq 1, \\ |h+v| - 1, & a+b > 1. \end{cases}$$

Let $Z(h)$ consist of the unit sphere, together with the point h if $|h| \leq 1$. Then

$$L(h, v) = \sup_{z \in Z(h)} \{z \cdot v + z \cdot (h - z)\}. \quad (6)$$

Indeed, the sphere supremum is $2a - 1$, the additional value is $a^2 - b^2$, and their difference is $(1 - a - b)(1 - a + b)$. If $|h| \leq 1 < a + b$, then $a - b \leq |h| \leq 1$, so the sphere value is the larger one. The other cases follow directly. In particular $L(h, v) \geq |h+v| - 1$ everywhere, and L is continuous.

Let Q be a flat torus of volume one. We claim that smooth periodic fields

$$h = h_0 + D\eta, \quad v = v_0 + \tilde{v}, \quad \operatorname{div} \tilde{v} = 0, \quad \int_Q \tilde{v} = 0$$

satisfy

$$\int_Q L(h, v) dx \geq L(h_0, v_0). \quad (7)$$

If $(|h_0 + v_0| + |h_0 - v_0|)/2 > 1$, this follows from $L(h, v) \geq |h+v| - 1$ and convexity of the norm. In the remaining case $|h_0| \leq 1$. Project η in $L^2(Q)$ onto the nonempty closed convex set

$$\mathcal{C} = \{\xi \in W^{1,\infty}(Q) : |h_0 + D\xi| \leq 1 \text{ a.e.}\}.$$

For completeness, this set is closed in L^2 : an L^2 -convergent sequence in $\mathcal{C}$ has uniformly bounded gradients and means; a subsequence converges uniformly, and weak-star convergence of its gradients preserves the convex gradient constraint. Denote the projection by ζ , and write

$$z = h_0 + D\zeta, \quad r = \eta - \zeta.$$

Its variational inequality is

$$\int_Q r(\xi - \zeta) dx \leq 0 \quad (\xi \in \mathcal{C}). \quad (8)$$

We have $z = h$ or $|z| = 1$ almost everywhere. To verify this assertion, put $w(x) = h_0 \cdot x + \zeta(x)$ locally. At a differentiability point x_0 where $r(x_0) > 0$ and $|Dw(x_0)| < 1$, choose a sufficiently small ball on which $r > 0$. A cone

$$c(x) = w(x_0) + \varepsilon - |x - x_0|$$

with sufficiently small $\varepsilon > 0$ lies below w on the boundary of that ball and above it at x_0 . The local replacement of w by $\max(w, c)$ preserves the gradient bound and gives a strictly positive left side in (8), a contradiction. At points where $r < 0$, use the upward cone and a minimum. On $\{r = 0\}$, $Dr = 0$ almost everywhere. Rademacher's theorem completes the assertion.

Translations of ζ are admissible. Applying (8) to its translations by $\pm te_i$, adding, and changing variables gives

$$\int_Q [r(x + te_i) - r(x)] [\zeta(x + te_i) - \zeta(x)] dx \geq 0.$$

Divide by t^2 , pass to the limit, and sum in i . Since $\int_Q Dr = 0$, the result is

$$\int_Q z \cdot (h - z) dx \geq 0. \quad (9)$$

Moreover, $z \in Z(h)$ almost everywhere and $\int_Q z \cdot v = h_0 \cdot v_0$ by the divergence-free condition. Equations (6) and (9) prove (7) in the remaining case.

3.2 A homogeneous Hessian calibration

Fix $1 < s < 2$. For $H \in \text{Sym}(3)$, define

$$N(H) = (\text{tr } H)I - H.$$

For $e \in S^2$ and $t > 0$, set

$$h_t(H) = \frac{e}{2} + \frac{He}{t}, \quad v_t(H) = \frac{N(H)e}{t}, \quad \ell_t(H) = \frac{\text{tr } H - e \cdot He}{2t},$$

and

$$F_e(H) = \int_0^\infty t^{s-1} [L(h_t(H), v_t(H)) - \ell_t(H)] dt, \quad F(H) = \frac{1}{4\pi} \int_{S^2} F_e(H) de. \quad (10)$$

These integrals converge absolutely, uniformly in e after normalizing $|H| = 1$. In fact, for $t \geq C_0|H|$, the pair (h_t, v_t) is in the inner branch, because $(e/2, 0)$ is strictly inside that branch. On this range,

$$L(h_t, v_t) - \ell_t = \frac{(\text{tr } H)(e \cdot He) - |He|^2}{t^2}.$$

For $0 < t < C_0|H|$, its absolute value is at most $C(1 + |H|/t)$. In the outer branch this is immediate; in the inner branch both vectors have norm at most one. Thus

$$\int_0^\infty t^{s-1} |L(h_t, v_t) - \ell_t| dt \leq C_s |H|^s. \quad (11)$$

Consequently F_e and F are continuous, positively s -homogeneous, and satisfy $|F(H)| \leq C_s |H|^s$, with $F(0) = 0$. Continuity away from zero follows by dominated convergence after normalization, and the growth bound gives continuity at zero.

For $H = H_0 + D^2\varphi$ with smooth periodic φ , the field h_t is a constant plus a scalar gradient, while v_t is divergence-free, since

$$\text{div } N(D^2\varphi) = 0.$$

Their means are $h_t(H_0)$, $v_t(H_0)$, and ℓ_t is linear in H . Apply (7), subtract the matching means of ℓ_t , and integrate with weight t^{s-1} . Fubini is justified by (11). This proves

$$\int_Q F(H_0 + D^2\varphi) dx \geq F(H_0). \quad (12)$$

The same proof with normalized spatial averages applies to a torus of any volume. In particular, placing a compactly supported test function in a sufficiently large torus gives

$$\int_{\mathbb{R}^3} F(D^2\varphi) dx \geq 0 \quad (\varphi \in C_c^\infty(\mathbb{R}^3)). \quad (13)$$

The definition and this inequality hold without any restriction on the sign of $\text{tr } H$.

3.3 An exact formula and angular convexity

Suppose now that $f = \text{tr } H \geq 0$. For a unit vector e , write

$$h = e \cdot He, \quad k = |He|^2, \quad A = f + h, \quad B = fh - k,$$

$$D = A^2 - 8B = (f - 3h)^2 + 8(k - h^2) \geq 0, \quad T = \frac{A + \sqrt{D}}{2}.$$

The inner branch condition for $L(h_t, v_t)$ is equivalent to

$$t \geq \frac{2f}{3}, \quad t^2 - At + 2B \geq 0. \quad (14)$$

Indeed, $h_t + v_t = (1/2 + f/t)e$, so the sum-of-lengths condition is

$$\left| \frac{t}{2}e + (2H - fI)e \right| \leq \frac{3t}{2} - f.$$

The right side must be nonnegative, and squaring then yields (14). At $t = 2f/3$ the polynomial is

$$-2[(h - f/3)^2 + k - h^2] \leq 0.$$

It follows that its outer root T satisfies $T \geq 2f/3$, and the inner branch occurs precisely for $t \geq T$, apart from equality endpoints. When $f = 0$, the same conclusion follows directly; $T = 0$ occurs only if $He = 0$.

For $0 < t < T$ the subtracted integrand in (10) is $A/(2t) - 1/2$; for $t > T$ it is B/t^2 . Direct integration gives

$$F_e(H) = -\frac{T^s}{2s} + \frac{AT^{s-1}}{2(s-1)} + \frac{BT^{s-2}}{2-s} \quad (15)$$

$$= \frac{T^{s-1}}{2(2-s)} \left(\frac{A}{s-1} - \frac{2T}{s} \right). \quad (16)$$

The second equality uses $T^2 - AT + 2B = 0$. At $T = 0$, the formula means its continuous value zero.

For fixed $f > 0$, introduce

$$I(\tau; A, B) = -\frac{\tau^s}{2s} + \frac{A\tau^{s-1}}{2(s-1)} + \frac{B\tau^{s-2}}{2-s}.$$

Its derivative is

$$I_\tau = -\frac{1}{2}\tau^{s-3}(\tau^2 - A\tau + 2B).$$

Since $2f/3$ lies between the roots of the polynomial, I increases up to T and decreases thereafter on $[2f/3, \infty)$. Therefore

$$F_e(H) = \sup_{\tau \geq 2f/3} I(\tau; A, B). \quad (17)$$

For $f = 0$ the same formula holds with the fixed domain $\tau > 0$: here $B = -k \leq 0$ and the other root is nonpositive. If $T = 0$, then $A = B = 0$, and the supremum is $\sup_{\tau > 0} (-\tau^s/(2s)) = 0$. Thus this supremum representation includes every degeneracy.

Diagonalize H , with eigenvalues $\lambda_1, \lambda_2, \lambda_3$, and write $w_i = e_i^2$ in an orthonormal eigenbasis. Then

$$h = \sum_i w_i \lambda_i, \quad k = \sum_i w_i \lambda_i^2, \quad w_i \geq 0, \quad \sum_i w_i = 1.$$

Both A and B are affine functions of w . In (17) the domain of τ depends only on the fixed trace f , and the function inside the supremum is affine in w . Hence $F_e(H)$ is convex as a function of these weights on their entire closed simplex. Let $V_f(\lambda_i)$ denote its value at the i th vertex. Convexity and $(4\pi)^{-1} \int_{S^2} e_i^2 de = 1/3$ imply

$$F(H) \leq \frac{1}{3} \sum_{i=1}^3 V_f(\lambda_i). \quad (18)$$

At a vertex, $h = \lambda$, $k = \lambda^2$, and

$$T = \max\{f - \lambda, 2\lambda\}. \quad (19)$$

3.4 Concavity of the vertex values and the sharp cone sign

Assume $3/2 \leq s < 2$. Define $W_f = 2(2-s)s(s-1)V_f$. Equations (16) and (19) give

$$W_f(\lambda) = \begin{cases} 2^{s-1}\lambda^{s-1}[sf + (4-3s)\lambda], & \lambda \geq f/3, \\ (f-\lambda)^{s-1}[(2-s)f + (3s-2)\lambda], & \lambda \leq f/3. \end{cases} \quad (20)$$

For $f > 0$, the second derivatives on the two branches are, respectively,

$$\begin{aligned} 2^{s-1}s(s-1)\lambda^{s-3}[-(2-s)f - (3s-4)\lambda] &< 0, \\ s(s-1)(f-\lambda)^{s-3}[-2(2-s)f - (3s-2)(f-\lambda)] &< 0. \end{aligned}$$

At $\lambda = f/3$, the values match, and both first derivatives are $s(2f/3)^{s-1}$. Thus V_f is globally strictly concave. For $f = 0$, the branches of W_0 are

$$2^{s-1}(4-3s)\lambda^s \quad (\lambda \geq 0), \quad -(3s-2)(-\lambda)^s \quad (\lambda \leq 0).$$

They are strictly concave, with matching derivative zero at the origin, so the conclusion holds also at zero trace.

Let $H \neq 0$ have trace $f \geq 0$ and smallest eigenvalue $-a < 0$. Put

$$m = \frac{\lambda_2 + \lambda_3}{2} = \frac{f+a}{2} > 0.$$

By concavity and (18),

$$F(H) \leq \frac{1}{3}[V_f(-a) + 2V_f(m)]. \quad (21)$$

Both vertex values on the right have transition $T = 2m$. Since $f = 2m - a$, their sum is exactly

$$V_f(-a) + 2V_f(m) = \frac{2(2m)^{s-1}}{(2-s)s(s-1)}[(3-s)m - sa]. \quad (22)$$

No positivity of λ_2 is required.

For $0 < \delta < 1/3$, write

$$P_\delta(H) = \delta \operatorname{tr} H + (1-3\delta)\lambda_{\min}(H) = \min_{\substack{A \geq \delta I \\ \operatorname{tr} A = 1}} A : H.$$

If $H \neq 0$, $f \geq 0$, and $P_\delta(H) \leq 0$, its smallest eigenvalue must be negative: a nonzero positive semidefinite matrix has $P_\delta(H) \geq \delta \operatorname{tr} H > 0$. The residual inequality becomes

$$2\delta m - (1-2\delta)a \leq 0, \quad \frac{a}{m} \geq \frac{2\delta}{1-2\delta}.$$

Since

$$\frac{2((3-s)/6)}{1-2((3-s)/6)} = \frac{3-s}{s},$$

equations (21)–(22) prove

$$F(H) < 0 \quad \text{if } H \neq 0, \quad \operatorname{tr} H \geq 0, \quad P_\delta(H) \leq 0, \quad \delta > \frac{3-s}{6}. \quad (23)$$

At the critical value $\delta = (3-s)/6$ the inequality is non-strict, with equality precisely on positive multiples and rotations of $\operatorname{diag}(-(3-s)/s, 1, 1)$. Indeed strict concavity forces the two larger eigenvalues to agree, and (22) then forces their critical ratio. For such a spectrum, $T(e) = 2m$ for every direction: $2m$ solves the quadratic for every directional weight, and the other root is $h-a \leq m-a < 2m$. Thus the angular reduction is an equality there. At zero trace and $H \neq 0$, $m = a/2$, so the bracket in (22) is $3(1-s)a/2 < 0$.

3.5 The Laplacian obstacle bridge

We now take $3/2 < s < 2$ and $\delta > (3-s)/6$. Continuity, homogeneity, and (23) imply constants $c, C > 0$, depending only on δ, s , such that

$$F(H) \leq C(P_\delta(H)_+)^s - c|H|^s \quad (\text{tr } H \geq 0). \quad (24)$$

To see this, restrict to the compact set $\{|H| = 1, \text{tr } H \geq 0\}$. The set where $P_\delta \leq 0$ is compact and F is strictly negative there, hence also on a neighborhood of that set. On the compact complement of this neighborhood, $(P_\delta)_+$ has a positive minimum. Choose C sufficiently large and use homogeneity.

First let u be C^2 on a neighborhood of $\overline{B_R}$, where $R \leq 1$. Set

$$b = \min_{\partial B_R} u, \quad M = (b - \min_{B_R} u)_+.$$

If $M = 0$ there is nothing to estimate. Otherwise the function

$$\psi(x) = \begin{cases} (b - u(x))_+, & x \in B_R, \\ 0, & x \notin B_R \end{cases}$$

is a nonnegative compactly supported Lipschitz function with maximum M .

Minimize $\int_{\mathbb{R}^3} |Dw|^2$ over

$$\{w \in \dot{H}^1(\mathbb{R}^3) : w \geq \psi \text{ a.e.}\}.$$

The homogeneous Sobolev space embeds in L^6 ; the admissible set is nonempty, closed, and convex. Thus a unique minimizer exists. Positive variations show that $-\Delta w \geq 0$. Truncation by M shows that $0 \leq w \leq M$.

We record continuity and decay, so that only local smooth-obstacle regularity will be needed. Let L_ψ be the Lipschitz constant of ψ . For a translation h , put $w_h = w(\cdot + h)$ and $\psi_h = \psi(\cdot + h)$, and let

$$a_h = (w - w_h - L_\psi|h|)_+.$$

Then $w - a_h \geq \psi$ and $w_h + a_h \geq \psi_h$. The two variational inequalities, added together, give $\int |Da_h|^2 \leq 0$. Reversing the translation proves that w has a globally L_ψ -Lipschitz representative. Since $w \in L^6$, this representative tends to zero at infinity: otherwise its Lipschitz bound produces infinitely many disjoint fixed-radius balls with a fixed positive lower bound, contradicting L^6 integrability. Variations off contact show that w is harmonic on $\{w > \psi\}$.

The nonzero nonnegative superharmonic function w is strictly positive everywhere. For example, the mean-value inequality at any hypothetical zero would force all its spherical averages, and then w itself, to vanish. The contact set

$$K = \{w = \psi\}$$

is therefore a compact subset of $B_R \cap \{\psi > 0\}$. It is nonempty, since $w = M$ at a point where $\psi = M$. Near K , the obstacle is the C^2 function $b - u$. Local obstacle regularity gives $w \in C^{1,1}$ near K ; see [2], in particular the local $C^{1,1}$ estimate for a $C^{1,1}$ obstacle. Outside K , w is harmonic. Covering K by finitely many such neighborhoods shows that

$$g = -\Delta w \in L^\infty(\mathbb{R}^3), \quad g \geq 0, \quad \text{supp } g \subset K.$$

These bounds are only regularity statements for the individual obstacle solution; their constants will not enter the estimate.

Put $v = -w$ and $\Gamma(x) = -1/(4\pi|x|)$. By decay and uniqueness of the harmonic remainder,

$$v = \Gamma * g, \quad \Delta v = g, \quad \|v\|_\infty = M.$$

In particular $D^2v \in L^s(\mathbb{R}^3)$ by the classical Newtonian singular-integral estimate. Let A be any measurable admissible coefficient field and set $f = A : D^2u$ in B_R . At almost every contact point, $v = u - b$ and $v \leq u - b$ locally, so

$$D^2v \leq D^2u, \quad P_\delta(D^2v) \leq A : D^2v \leq f.$$

Off K , the trace of D^2v is zero, and hence $P_\delta(D^2v) \leq 0$. Consequently

$$\operatorname{tr} D^2v \geq 0, \quad (P_\delta(D^2v))_+ \leq f_+ \mathbf{1}_K \quad \text{a.e.} \quad (25)$$

The compact-test inequality (13) extends to v . Indeed, its potential representation gives

$$v = O(|x|^{-1}), \quad Dv = O(|x|^{-2}), \quad D^2v = O(|x|^{-3})$$

at infinity. Smooth cutoffs on expanding balls approximate v in Hessian L^s for $s > 1$, and subsequent smoothing gives compactly supported smooth approximants. The continuity and s -growth of F imply continuity of its integral under this L^s convergence, by convergence in measure and uniform integrability. Thus $\int F(D^2v) \geq 0$. Integrating (24) and using (25) yields

$$c\|D^2v\|_{L^s(\mathbb{R}^3)}^s \leq C\|f_+\|_{L^s(B_R)}^s.$$

Since $g = \operatorname{tr} D^2v$, we have $\|g\|_s \leq \sqrt{3}\|D^2v\|_s$. The Newtonian kernel belongs locally to $L^{s'}$ precisely when $s > 3/2$, where $s' = s/(s-1)$. Its convolution with a function supported in B_R , $R \leq 1$, therefore gives

$$M = \|v\|_\infty \leq C_s\|g\|_s \leq C_{\delta,s}\|f_+\|_{L^s(B_R)}. \quad (26)$$

The constants are uniform in $R \leq 1$. Applying the same argument to $-u$ bounds its positive excursion by $C_{\delta,s}\|f_-\|_{L^s(B_R)}$.

For the original class $u \in C^2(B_1) \cap C(\overline{B_1})$ with zero boundary data, apply these estimates on B_R and let $R \uparrow 1$. The boundary extrema tend uniformly to zero, while the source norms do not exceed the norm on B_1 . We obtain

$$\|u\|_{L^\infty(B_1)} \leq C_{\delta,s}\|A : D^2u\|_{L^s(B_1)} \quad \text{if } \frac{3}{2} < s < 2, \quad s > 3(1-2\delta). \quad (27)$$

No regularity of A beyond measurability was used.

3.6 Selection of the exponent

Let $1/6 < \delta < 1/3$ and let $p < \infty$ satisfy

$$p > \max \left\{ \frac{3}{2}, 3(1-2\delta) \right\}.$$

Since $3(1-2\delta) < 2$, one may choose

$$\max \left\{ \frac{3}{2}, 3(1-2\delta) \right\} < s < \min\{p, 2\}.$$

Equation (27) applies at this s , and Hölder's inequality on B_1 gives

$$\|u\|_{L^\infty(B_1)} \leq C_{\delta,s}|B_1|^{1/s-1/p}\|A : D^2u\|_{L^p(B_1)}.$$

Thus the asserted sufficiency holds throughout $1/6 < \delta < 1/3$, including target exponent $p = 2$, without taking a limit of the calibration as $s \uparrow 2$.

4 Sufficiency in the low ellipticity range

We prove that, if $0 < \delta \leq 1/6$ and $p > 3(1 - 2\delta)$, then

$$\|u\|_{L^\infty(B_1)} \leq C(\delta, p) \|A : D^2 u\|_{L^p(B_1)} \quad (28)$$

for every measurable symmetric matrix field A satisfying $\operatorname{tr} A = 1$ and $A \geq \delta I$, and every $u \in C^2(B_1) \cap C^0(\overline{B_1})$ with $u = 0$ on ∂B_1 . Set

$$\lambda = 1 - 2\delta, \quad q_* = \frac{3\lambda}{2(\lambda - \delta)} = \frac{3(1 - 2\delta)}{2 - 6\delta}.$$

In the present range, $3/2 < q_* \leq 2$. We first fix

$$\frac{3}{2} < q < q_*, \quad p = \frac{q}{q - 1}, \quad (29)$$

so that $3(1 - 2\delta) < p < 3$ and $2 < p < 3$.

Throughout this section we use the fixed flat torus

$$Q = \mathbb{R}^3 / (4\mathbb{Z})^3, \quad V = |Q| = 64, \quad \langle f \rangle = \frac{1}{V} \int_Q f(x) dx.$$

The ball B_1 lies inside a coordinate chart of Q . For a symmetric tensor T , write

$$\operatorname{div} \operatorname{div} T = \sum_{i,j=1}^3 \partial_i \partial_j T_{ij}, \quad \mathcal{C}_\delta = \{T = T^\top : \operatorname{tr} T \geq 0, T \geq \delta(\operatorname{tr} T)I\}.$$

This is a convex cone of positive semidefinite matrices.

4.1 A periodic determinant inequality

Let U be a smooth positive definite periodic tensor satisfying $\operatorname{div} \operatorname{div} U = 0$, and put

$$c = \langle \det(U)^{1/2} \rangle, \quad M = \langle U \rangle, \quad H_0 = \det(M)^{1/3} M^{-1}.$$

Then $\det H_0 = 1$. The periodic Monge–Ampère corrector theorem [1, Theorem 0.2], after rescaling the period, provides a smooth periodic function ψ such that

$$H = H_0 + D^2 \psi > 0, \quad \det H = \frac{\det(U)^{1/2}}{c}.$$

Indeed the prescribed right-hand side is smooth, positive, and has spatial average one, equal to $\det H_0$; the cited theorem permits any prescribed positive definite H_0 with this determinant. Matrix arithmetic–geometric mean gives

$$U : H \geq 3(\det U \det H)^{1/3} = 3c^{-1/3} \det(U)^{1/2}.$$

Twice integrating by parts on the torus yields

$$\langle U : D^2 \psi \rangle = \langle \psi \operatorname{div} \operatorname{div} U \rangle = 0.$$

Consequently

$$3c^{2/3} \leq \langle U : H \rangle = M : H_0 = 3 \det(M)^{1/3},$$

and hence

$$\langle \det(U)^{1/2} \rangle \leq \det(\langle U \rangle)^{1/2}. \quad (30)$$

The argument uses only the double-divergence equation.

For a positive semidefinite matrix T , define

$$\Phi_q(T) = \int_0^\infty \left[\det(T + sI)^{1/2} - s^{3/2} - \frac{\text{tr } T}{2} s^{1/2} \right] s^{q-5/2} ds. \quad (31)$$

The bracket is bounded as $s \downarrow 0$. As $s \rightarrow \infty$, it has the expansion

$$\det(T + sI)^{1/2} - s^{3/2} - \frac{\text{tr } T}{2} s^{1/2} = \frac{(\text{tr } T)^2 - 2 \text{tr}(T^2)}{8} s^{-1/2} + O(s^{-3/2}). \quad (32)$$

These bounds are uniform when T ranges over a bounded set of positive semidefinite matrices. Thus (31) converges absolutely for $3/2 < q < 2$. Substitution $s = at$ shows $\Phi_q(aT) = a^q \Phi_q(T)$ for $a > 0$.

Suppose now that T is a smooth periodic positive semidefinite tensor with $\text{div div } T = 0$. For every $s > 0$, the tensor $T + sI$ is positive definite and satisfies the same equation. Apply (30) to $T + sI$, subtract the affine terms in (31), and integrate against the positive weight $s^{q-5/2}$ on a finite interval. The affine terms have matching spatial averages. Absolute convergence, uniformly in the spatial variable, then gives

$$\langle \Phi_q(T) \rangle \leq \Phi_q(\langle T \rangle). \quad (33)$$

4.2 The exact spectral coercivity

We evaluate (31) on $T = \text{diag}(a, b, b)$, where $a > 0$ and $b \geq 0$. Since $\det(T + sI)^{1/2} = (s + b)\sqrt{s + a}$, we have

$$\Phi_q(\text{diag}(a, b, b)) = I_1(a) + bI_2(a),$$

where

$$\begin{aligned} I_1(a) &= \int_0^\infty s^{q-3/2} \left(\sqrt{s + a} - \sqrt{s} - \frac{a}{2\sqrt{s}} \right) ds, \\ I_2(a) &= \int_0^\infty s^{q-5/2} \left(\sqrt{s + a} - \sqrt{s} \right) ds. \end{aligned}$$

Both integrals converge absolutely. They are homogeneous in a of degrees q and $q - 1$, respectively. Write

$$B_q = B\left(q - \frac{3}{2}, 2 - q\right) > 0, \quad B(x, y) = \int_0^\infty t^{x-1} (1 + t)^{-x-y} dt.$$

Differentiation under the integral sign, followed by $s = at$, gives

$$\begin{aligned} I_2'(a) &= \frac{1}{2} \int_0^\infty s^{q-5/2} (s + a)^{-1/2} ds = \frac{B_q}{2} a^{q-2}, \\ I_1''(a) &= -\frac{1}{4} \int_0^\infty s^{q-3/2} (s + a)^{-3/2} ds \\ &= -\frac{1}{4} B\left(q - \frac{1}{2}, 2 - q\right) a^{q-2} = -\frac{(2q - 3)B_q}{4} a^{q-2}. \end{aligned}$$

The differentiations are justified locally uniformly for $a > 0$ by the same endpoint estimates. The last identity uses $B(x + 1, y) = xB(x, y)/(x + y)$. Homogeneity determines the constants without additional affine terms:

$$I_2(a) = \frac{B_q}{2(q - 1)} a^{q-1}, \quad I_1(a) = -\frac{(2q - 3)B_q}{4q(q - 1)} a^q.$$

Therefore, with $k_q = B_q/[4q(q-1)]$,

$$\Phi_q(\text{diag}(a, b, b)) = k_q[(3-2q)a^q + 2qb a^{q-1}]. \quad (34)$$

Let $T \in \mathcal{C}_\delta$ and $\rho = \text{tr } T > 0$. Its eigenvalues belong to the simplex

$$t_i \geq \delta\rho, \quad t_1 + t_2 + t_3 = \rho,$$

whose vertices are the permutations of $(\lambda\rho, \delta\rho, \delta\rho)$. For every $s > 0$, concavity of $\sum_i \log(s + t_i)$ implies

$$\det(T + sI) \geq (s + \lambda\rho)(s + \delta\rho)^2.$$

The trace subtractions in (31) are identical at fixed trace. It follows from (34) that

$$\Phi_q(T) \geq k_q \lambda^{q-1} [3\lambda - 2q(\lambda - \delta)] \rho^q. \quad (35)$$

The inequality also holds at $T = 0$. Conversely, arithmetic–geometric mean at fixed trace gives, for every positive semidefinite S ,

$$\det(S + sI) \leq \left(s + \frac{\text{tr } S}{3}\right)^3,$$

and hence

$$\Phi_q(S) \leq k_q 3^{1-q} (\text{tr } S)^q. \quad (36)$$

The bracket in (35) is positive exactly when $q < q_*$. Combining (33), (35), and (36) proves

$$\|\text{tr } T\|_{L^q(Q)} \leq K V^{1/q-1} \int_Q \text{tr } T, \quad K^q = \frac{(3\lambda)^{1-q}}{3\lambda - 2q(\lambda - \delta)}. \quad (37)$$

Indeed the corresponding inequality for spatial averages is $\langle \rho^q \rangle \leq K^q \langle \rho \rangle^q$. This fixes the volume normalization in (37).

4.3 A positive compensator

For $x \neq 0$ in a coordinate ball of Q , set

$$\mathcal{R}_0(x) = \frac{c_\delta}{|x|} [\lambda n \otimes n + \delta(I - n \otimes n)], \quad n = \frac{x}{|x|}, \quad c_\delta = \frac{1}{4\pi(1-4\delta)}.$$

Here $c_\delta > 0$ because $\delta \leq 1/6$. The trace of $\mathcal{R}_0$ is $c_\delta/|x|$, and $\mathcal{R}_0 \in \mathcal{C}_\delta$. Direct differentiation gives

$$\text{Div } \mathcal{R}_0 = c_\delta(1-4\delta) \frac{x}{|x|^3}, \quad \text{div div } \mathcal{R}_0 = \delta_0 \quad \text{in distributions.}$$

The first divergence has no additional point mass: the boundary term from $\mathcal{R}_0$ on a sphere of radius r is $O(r)$ against a bounded test function. The second identity follows from $\text{div}(x/|x|^3) = 4\pi\delta_0$.

Choose $0 < \varepsilon < 1$ and a smooth cutoff $0 \leq \chi_\varepsilon \leq 1$ that equals one near zero and is supported in B_ε . Then

$$\text{div div}(\chi_\varepsilon \mathcal{R}_0) = \delta_0 + f_\varepsilon, \quad \int_Q f_\varepsilon = -1,$$

where f_ε is smooth and supported in the cutoff annulus. The smooth periodic Poisson equation

$$\Delta h_\varepsilon = -\frac{1}{V} - f_\varepsilon$$

is solvable because its right-hand side has integral zero. Choose a constant k_ε so that $h_\varepsilon + k_\varepsilon \geq 0$, and put

$$H_\varepsilon = (h_\varepsilon + k_\varepsilon)I, \quad \mathcal{R} = \chi_\varepsilon \mathcal{R}_0 + H_\varepsilon.$$

Then

$$\mathcal{R} \in \mathcal{C}_\delta, \quad \operatorname{div} \operatorname{div} \mathcal{R} = \delta_0 - \frac{1}{V}. \quad (38)$$

Define

$$M_\varepsilon = \|H_\varepsilon\|_{L^\infty(Q; \operatorname{op})} < \infty, \quad a_\varepsilon = \|\chi_\varepsilon \mathcal{R}_0\|_{L^p(Q; \operatorname{op})}.$$

Since $p < 3$ and $\|\mathcal{R}_0(x)\|_{\operatorname{op}} \leq c_\delta/|x|$,

$$a_\varepsilon \leq c_\delta \left(\frac{4\pi}{3-p} \right)^{1/p} \varepsilon^{3/p-1} \longrightarrow 0 \quad (\varepsilon \downarrow 0). \quad (39)$$

The finite constant M_ε may grow as ε decreases; once ε is fixed, it depends only on δ, ε and Q .

4.4 The periodic resolvent estimate

Let A be smooth and periodic, with $\operatorname{tr} A = 1$ and $A \geq \delta I$, and let $g \geq 0$ be smooth with $\int_Q g = 1$. Write $L_A u = A : D^2 u$ and let

$$G = (1 - L_A^*)^{-1} g.$$

Smooth periodic elliptic resolvent theory gives a smooth G . The maximum principle makes the forward resolvent positive, so duality gives $G \geq 0$. Testing the adjoint equation against the constant function gives

$$\operatorname{div} \operatorname{div}(AG) = G - g, \quad \int_Q G = 1. \quad (40)$$

In particular $\Gamma = \|G\|_{L^q(Q)}$ is finite for each fixed smooth A, g .

Convolution on Q is taken with unnormalized Lebesgue measure: $(\mathcal{R} * G)(x) = \int_Q \mathcal{R}(x - y)G(y) dy$. Hölder's inequality and (39) imply

$$\|\mathcal{R} * G\|_{L^\infty(Q; \operatorname{op})} \leq a_\varepsilon \Gamma + M_\varepsilon.$$

Set

$$\kappa = \frac{a_\varepsilon \Gamma + M_\varepsilon}{1 - 3\delta}, \quad \mathcal{T} = AG + \mathcal{R} * g - \mathcal{R} * G + \kappa I.$$

Both convolutions are smooth because $\mathcal{R} \in L^1(Q)$ and g, G are smooth. Equations (38) and (40) give

$$\operatorname{div} \operatorname{div} \mathcal{T} = (G - g) + (g - V^{-1}) - (G - V^{-1}) = 0.$$

The convexity of $\mathcal{C}_\delta$ and positivity of g imply $\mathcal{R} * g \in \mathcal{C}_\delta$. Moreover $\mathcal{R} * G$ is positive semidefinite, and

$$\begin{aligned} & (\kappa I - \mathcal{R} * G) - \delta \operatorname{tr}(\kappa I - \mathcal{R} * G)I \\ &= (1 - 3\delta)\kappa I - \mathcal{R} * G + \delta \operatorname{tr}(\mathcal{R} * G)I \geq 0. \end{aligned}$$

The chosen κ also dominates the operator norm of $\mathcal{R} * G$. Thus $\kappa I - \mathcal{R} * G \in \mathcal{C}_\delta$, and consequently

$$\mathcal{T} \in \mathcal{C}_\delta, \quad \operatorname{tr} \mathcal{T} \geq G.$$

Since $\int_Q g = \int_Q G = 1$, the convolved terms cancel in the integral:

$$\int_Q \text{tr } \mathcal{T} = 1 + 3\kappa V.$$

Applying (37) therefore yields

$$\Gamma \leq KV^{1/q-1} \left[1 + \frac{3V}{1-3\delta} (a_\varepsilon \Gamma + M_\varepsilon) \right].$$

Choose ε , depending only on δ, q , so small that

$$\frac{3KV^{1/q}a_\varepsilon}{1-3\delta} \leq \frac{1}{2}.$$

Absorbing the term containing Γ gives

$$\|G\|_{L^q(Q)} \leq 2KV^{1/q-1} \left[1 + \frac{3VM_\varepsilon}{1-3\delta} \right] =: C_1(\delta, q). \quad (41)$$

This constant is independent of A and g . Smoothness supplied the finiteness of Γ before absorption, so no preliminary uniform Green kernel estimate is required.

For smooth periodic F and $v = (1 - L_A)^{-1}F$, duality now gives

$$\left| \int_Q vg \right| = \left| \int_Q FG \right| \leq C_1 \|F\|_{L^p(Q)}$$

for every smooth probability density g . Letting such densities concentrate at any prescribed point, and using continuity of v , proves

$$\|(1 - L_A)^{-1}F\|_{L^\infty(Q)} \leq C_1 \|F\|_{L^p(Q)}. \quad (42)$$

4.5 Comparison and passage to measurable coefficients

Fix $0 < R \leq 1$ and first suppose that u is C^2 on a neighborhood of $\overline{B_R}$. Set

$$B = \sup_{\partial B_R} |u|, \quad M = (\|u\|_{L^\infty(B_R)} - B)_+, \quad b_R(x) = \frac{R^2 - |x|^2}{2}.$$

For each $\eta > 0$ choose a nonnegative smooth periodic F_η such that

$$F_\eta \geq |L_A u| \quad \text{on } B_R, \quad \|F_\eta\|_{L^p(Q)} \leq \|L_A u\|_{L^p(B_R)} + \eta.$$

To obtain these majorants, extend the continuous function $|L_A u|$ to a thin collar of the closed ball, cut it off, approximate uniformly by smooth functions, and add a vanishing positive constant. The collar contribution tends to zero in L^p . Let

$$v_\eta = (1 - L_A)^{-1}F_\eta \geq 0, \quad w_\eta = B + v_\eta + Mb_R.$$

The trace normalization gives

$$(1 - L_A)b_R = 1 + b_R.$$

Since $B + M \geq |u|$ and $F_\eta \geq |L_A u|$,

$$\begin{aligned} (1 - L_A)(w_\eta - u) &= B + F_\eta + M(1 + b_R) - u + L_A u \geq 0, \\ (1 - L_A)(w_\eta + u) &= B + F_\eta + M(1 + b_R) + u - L_A u \geq 0. \end{aligned}$$

Also $w_\eta \geq |u|$ on ∂B_R . The maximum principle implies $|u| \leq w_\eta$, and hence

$$\|u\|_{L^\infty(B_R)} \leq B + \|v_\eta\|_{L^\infty(Q)} + \frac{M}{2}.$$

Using (42) and then letting $\eta \downarrow 0$ gives the uniform estimate

$$\|u\|_{L^\infty(B_R)} \leq \sup_{\partial B_R} |u| + 2C_1 \|L_A u\|_{L^p(B_R)}. \quad (43)$$

Now let A be merely measurable on B_1 , and let $u \in C^2(B_1) \cap C^0(\overline{B_1})$ vanish on ∂B_1 . Extend A to Q by $I/3$ outside B_1 and convolve periodically. The resulting smooth fields A_j retain symmetry, trace one, and the inequality $A_j \geq \delta I$, and converge almost everywhere to A in B_1 . For each fixed $R < 1$, the Hessian $D^2 u$ is bounded on $\overline{B_R}$. Thus dominated convergence gives

$$A_j : D^2 u \longrightarrow A : D^2 u \quad \text{in } L^p(B_R).$$

Apply (43) to A_j and pass to the limit $j \rightarrow \infty$. Then let $R \uparrow 1$. Continuity of u on the closed ball and its zero boundary values give $\sup_{\partial B_R} |u| \rightarrow 0$, proving (28) for $3(1 - 2\delta) < p < 3$.

Finally, if $p \geq 3$, choose p_0 with $3(1 - 2\delta) < p_0 < 3 \leq p$ and use

$$\|A : D^2 u\|_{L^{p_0}(B_1)} \leq |B_1|^{1/p_0 - 1/p} \|A : D^2 u\|_{L^p(B_1)}.$$

This proves (28) for every $p > 3(1 - 2\delta)$, without any assumption on the global integrability of $D^2 u$.

5 Completion of the equivalence

If $0 < \delta \leq 1/6$, the preceding adjoint argument proves (1) for every $p > 3(1 - 2\delta) = p_c(\delta)$. If $1/6 < \delta < 1/3$ and $p > p_c(\delta)$, choose

$$s = \frac{p_c(\delta) + \min\{p, 2\}}{2}.$$

Then $3/2 < s < 2$, $s > 3(1 - 2\delta)$ and $s < p$. The directional calibration and obstacle argument prove the estimate at exponent s . Hölder's inequality on B_1 gives

$$\|L_A u\|_{L^s(B_1)} \leq |B_1|^{1/s - 1/p} \|L_A u\|_{L^p(B_1)},$$

with s determined solely by δ, p . This includes $p = 2$ whenever it lies above the threshold; no limiting estimate at exponent two is needed. The counterexamples in Section 2 prove the converse and exclude equality at either critical exponent. This proves Theorem 1.1 in its exact stated class. $\square$

References

- [1] L. Caffarelli and Y. Y. Li, *A Liouville theorem for solutions of the Monge–Ampère equation with periodic data*, Annales de l’Institut Henri Poincaré C, Analyse non linéaire **21** (2004), 97–120. Theorem 0.2, p. 99. [Primary PDF](#).
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